

COMP-3250 Midterm 2 Cheat Sheet

Gradient Boosting Machines (GBM)

- Ensemble method that builds models sequentially.
- Each model corrects errors from the previous one.
- Uses gradient descent to minimize a loss function.
- Common params: `n_estimators`, `learning_rate`, `max_depth`, `subsample`.

Kernelized Support Vector Machines (SVMs)

- SVMs find decision boundaries maximizing margin between classes.
- Kernel trick maps data into higher dimensions implicitly.
- Common kernels: linear, polynomial, RBF (Gaussian).
- scikit-learn: `SVC(kernel='rbf', C=1.0, gamma='scale')`

Multilayer Perceptrons (MLP)

- Feedforward neural networks with input, hidden, and output layers.
- Each neuron applies weights, bias, and activation function.
- Common activations: ReLU, sigmoid, tanh.
- Trained with backpropagation and gradient descent.

Principal Component Analysis (PCA)

- Unsupervised method for dimensionality reduction.
- Finds orthogonal axes (principal components) of max variance.
- Requires standardized data (mean=0, variance=1).
- scikit-learn: `PCA(n_components=2)`